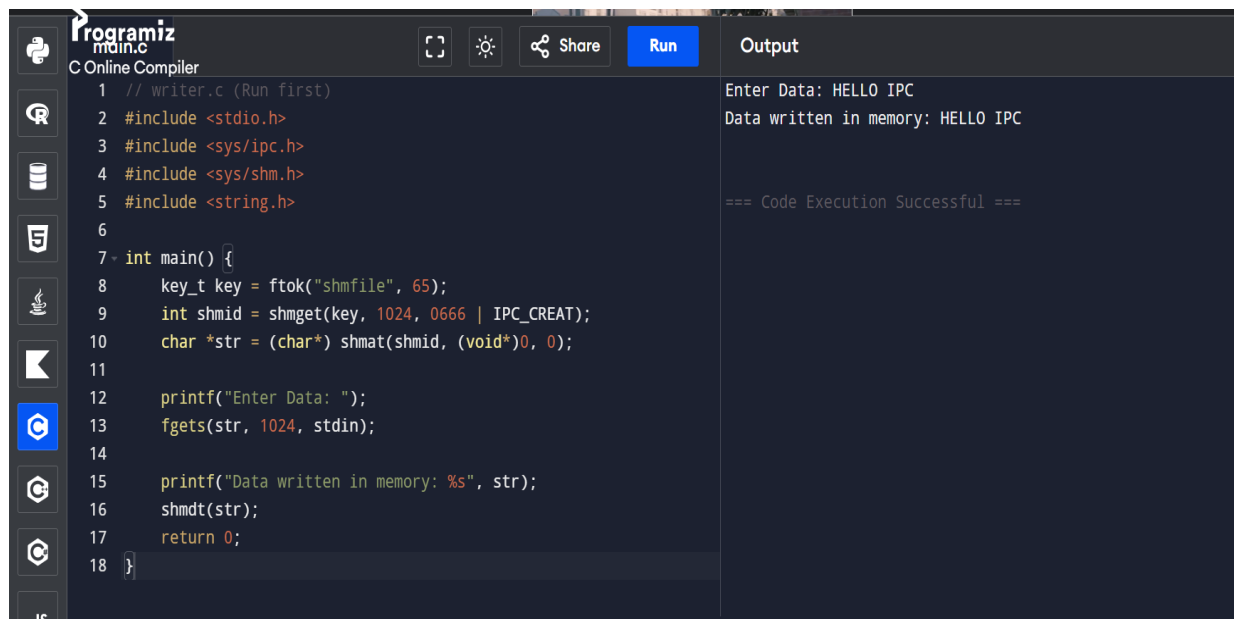


EXPERIMENT 09:

9. Illustrate the concept of inter-process communication using shared memory with a C program.



```
1 // writer.c (Run first)
2 #include <stdio.h>
3 #include <sys/ipc.h>
4 #include <sys/shm.h>
5 #include <string.h>
6
7 int main() {
8     key_t key = ftok("shmfile", 65);
9     int shmid = shmget(key, 1024, 0666 | IPC_CREAT);
10    char *str = (char*) shmat(shmid, (void*)0, 0);
11
12    printf("Enter Data: ");
13    fgets(str, 1024, stdin);
14
15    printf("Data written in memory: %s", str);
16    shmdt(str);
17    return 0;
18 }
```

Output

Enter Data: HELLO IPC
Data written in memory: HELLO IPC

=== Code Execution Successful ===

PSEUDO CODE :

PROCESS 1 (Writer):

1. Create/Get shared memory segment using shmget().
2. Attach to the memory segment using shmat().
3. Prompt user for string input and copy it into shared memory.
4. Detach memory using shmdt().

PROCESS 2 (Reader):

1. Get shared memory segment using shmget().
2. Attach to memory segment using shmat().
3. Read and print string from shared memory.
4. Detach and destroy memory segment using shmctl().

CODE:

```
// writer.c (Run first)

#include <stdio.h>

#include <sys/ipc.h>

#include <sys/shm.h>

#include <string.h>


int main() {

    key_t key = ftok("shmfile", 65);

    int shmid = shmget(key, 1024, 0666 | IPC_CREAT);

    char *str = (char*) shmat(shmid, (void*)0, 0);


    printf("Enter Data: ");

    fgets(str, 1024, stdin);


    printf("Data written in memory: %s", str);

    shmdt(str);

    return 0;

}


// reader.c (Run second)

#include <stdio.h>

#include <sys/ipc.h>

#include <sys/shm.h>


int main() {

    key_t key = ftok("shmfile", 65);

    int shmid = shmget(key, 1024, 0666);
```

```
char *str = (char*) shmat(shmid, (void*)0, 0);

printf("Data read from memory: %s", str);

shmdt(str);

shmctl(shmid, IPC_RMID, NULL); // Destroy shared memory

return 0;
}
```